ĐỀ THI KẾT THÚC HỌC PHẦN
Học kỳ 2 – Năm học 2022-2023

MÃ LƯU TRỮ
(*do phòng KT-ĐBCL ghi*)

Tên học phần:	**Applied Statistics for Engineering and Scientists 1**	**Mã HP:**	
Thời gian làm bài:	**120'**	Ngày thi:	

Ghi chú: *Sinh viên* [☒ ***được phép sử dụng tài liệu giấy*** / ☐ *không được phép*]

Họ tên sinh viên: ….. **MSSV**: …………… **STT**: …..

Question 1. (1.5 points)

In 2002, the average height of a woman aged 20–74 years was 64 inches with an increase of approximately 1 inch from 1960 (http://usgovinfo.about.com/od/healthcare). Suppose the height of a woman is normally distributed with a standard deviation of two inches.

(a) What is the probability that a randomly selected woman in this population is between 58 inches and 70 inches?

(b) Determine the height that is symmetric about the mean that includes 90% of this population.

(c) What is the probability that two of five women selected at random from this population exceed 68 inches?

Question 2. (1.5 points)

A geneticist is interested in the proportion of African males who have a certain minor blood disorder. In a random sample of 100 African males, 24 are found to be afflicted. Compute a 99% confidence interval for the proportion of African males who have this blood disorder.

Question 3. (2.0 points)

The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint:

3.4, 2.5, 4.8, 2.9, 3.6, 2.8, 3.3, 5.6, 3.7, 2.8, 4.4, 4.0, 5.2, 3.0, 4.8

(a) Assuming that the measurements represent a random sample from a normal population, find a 95% confidence interval for the drying time for the next trial of the paint.

(b) What sample size is required so that we have 95% confidence that the maximum error of the estimate of μ is 0.2?

Question 4. (1.5 points)

It is claimed that automobiles are driven on average more than 20,000 kilometers per year. To test this claim, 100 randomly selected automobile owners are asked to keep a record of the kilometers they travel. Would you agree with this claim if the random sample showed an average of 23,500 kilometers and a standard deviation of 3900 kilometers? Use a P -value in your conclusion and $\alpha = 0.03$.

Question 5. (1.5 points)

A new radar device is being considered for a certain missile defense system. The system is checked by experimenting with aircraft in which a kill or a no kill is simulated. If, in 300 trials, 250 kills occur, accept or reject, at the 0.04 level of significance, the claim that the probability of a kill with the new system does not exceed the 0.8.

Question 6. (2.0 points)

In semiconductor manufacturing, wet chemical etching is often used to remove silicon from the backs of wafers prior to metallization. The etch rate is an important characteristic in this process and known to follow a normal distribution. Two different etching solutions have been compared using two random samples of 10 wafers for each solution. The observed etch rates are as follows (in mils per minute):

Solution 1		Solution 2	
9.9	10.6	10.2	10.0
9.4	10.3	10.6	10.2
9.3	10.0	10.7	10.7
9.6	10.3	10.4	10.4
10.2	10.1	10.5	10.3

(a) Do the data support the claim that the mean etch rate is the same for both solutions? In reaching your conclusions, use $\alpha = 0.03$ and assume that both population variances are equal.

(b) Find a 95% confidence interval on the difference in mean etch rates.